Algorithmic and Minimax Complexities in Kernel Bandits

Yunbei Xu
National University of Singapore
yunbei@nus.edu.sg

Abstract

This paper uses the model-index algorithmic information ratio (MAIR) to give a unified account of three central strategies for kernel bandits: GP–UCB, (GP–)E2D, and MAMS/MEBO. The starting point is that the MAIR gradient-bracket theorem of Xu and Zeevi (2025) is an exact identity before any relaxation: regret separates into a telescoping estimation-complexity term and a regret-to-information coefficient, providing a principled language for comparing algorithms that otherwise come from different traditions. Under this lens, GP–UCB and GP–E2D share the same algorithmic Gaussian-process posterior and hence the shared realized log-determinant estimation geometry. They differ on the coefficient side: GP–UCB controls the coefficient indirectly through calibration and optimism, whereas GP–E2D optimizes a DEC offset directly. MAMS/MEBO optimizes the same MAIR bracket in a robust way and naturally extends to pathwise adversarial environments, but finite-covering implementations typically lose the realized GP information-gain geometry. A hub–cloud construction illustrates why full-class minimax and DEC certificates can be pessimistic relative to the localized performance of a heterogeneous GP prior. The resulting view complements the intuition that GP-UCB need not be coefficient-optimal in the step-uniform minimax sense; that GP-E2D is designed precisely to address this gap; and that realized algorithmic trajectories should remain central to the long-standing debate on kernel-bandit optimality.[1]

Contents

[1] The current version represents the last full pass; new outlook can be found in appendix.

1 Introduction

Kernal and Gaussian-process bandits are a useful meeting point for two cultures of sequential learning. The first culture is algorithmic and practical. GP–UCB maintains a kernel ridge-regression posterior mean and variance, then queries the point with the largest upper confidence bound (Srinivas et al., 2010). GP–UCB and its variants have became influential because they are simple, reliable, and easy to implement in AI and decision-making applications. The second culture is information-theoretic and minimax. The DMSO framework and the decision-estimation coefficient (DEC) characterize interactive learning through an offset between decision regret and statistical information (Foster et al., 2021, 2023; Chen et al., 2024). Algorithms such as E2D and MAMS (also interpretable as model-index EBO) solve minimax versions of this offset (Foster et al., 2021; Xu and Zeevi, 2025; Lattimore and György, 2021).

The standard frequentist kernel-bandit model is not literally Bayesian. The truth is a fixed function $f^\star$ in an RKHS and the observation noise is conditionally sub-Gaussian. The Gaussian process used by GP–UCB is an algorithmic reference: it defines a posterior mean, a posterior variance, and a log-determinant information budget. This is why we use the phrase *kernel/Gaussian-process bandits*. The statistical model is best called a kernel or RKHS bandit; the algorithmic reference is Gaussian-process shaped.

The central claim of the paper is that GP–UCB, E2D, and MAMS are three projections of an exact MAIR regret identity. This identity makes rigorous a separation between the information gain and the regret-to-information coefficient. Kernel bandits provide a clean setting in which the resulting distinction between algorithmic information and class-wide minimax coefficients can be made mathematically precise.

This perspective helps clarify the long discussion around the information-theoretical optimality of GP–UCB. The original GP–UCB analysis of Srinivas et al. (2010) bounds regret by maximal information gain. Later work sharpened information-gain estimates and raised the problem of tight online confidence multipliers for RKHS elements (Vakili et al., 2021b,a). Whitehouse et al. (2023) showed that improved smoothness-aware regularization and Hilbert-space self-normalized concentration yield nearly optimal, and in particular sublinear, GP–UCB rates for important kernels such as Matérn kernels. In the Bayesian GP setting, Iwazaki (2025); Takeno and Iwazaki (2026) further emphasize that the realized information gained by the GP–UCB trajectory can be much smaller than the maximum-variance-reduction trajectory. On the other hand, Lattimore (2023) gives lower-bound evidence that fully general adaptive confidence intervals cannot be arbitrarily tightened, and Wang and Zhang (2026) gives a Matérn lower-bound obstruction for GP–UCB under polynomial effective optimism.

Starting from finite or covering-based model classes, DEC (Foster et al., 2021) and MAIR (Xu and Zeevi, 2025) provide robust, class-wide approaches to minimax sequential learning through regret-to-information tradeoffs. In this setting, DEC gives not only algorithmic upper bounds but also lower-bound certificates for the minimax regret–estimation coefficient (Foster et al., 2021, 2023). By contrast, GP–UCB is not automatically justified on the DEC/MAIR coefficient side: optimism bounds regret through confidence widths, but it does not solve the minimax offset problem. It is worth noting that Kirschner and Krause (2018) proposed frequentist GP-IDS and showed that its information-ratio certificate can improve upon the GP-UCB certificate on a per-step basis, although no strict multi-step consequence was established. The identity-based separation in this paper clarifies why this improvement is structural: GP–UCB and GP–E2D/GP–IDS can share the same GP log-determinant estimation geometry, while GP–E2D optimizes the regret-to-information coefficient through a class-wide minimax decision rule to optimize DEC. Together with the step-uniform minimax optimality of DEC, this suggests further insight against a strengthened coefficient-side optimality conjecture for GP-UCB: UCB confidence multiplier need not be *step-uniform* minimax optimal as a regret-to-information conversion coefficient, while E2D is designed to address precisely this gap; see Section 7.2 for the detailed background.

At the same time, finite-covering implementations of E2D and MAMS can be pessimistic for kernel problems. Their covering and discretization costs may dominate the clean log-determinant estimation complexity that makes GP–UCB practical. The unified MAIR viewpoint extends all three algorithms and clarifies their relative advantages. In particular, it identifies the Gaussian-process posterior as the natural estimation oracle for E2D: the resulting GP–E2D algorithm retains the GP information-gain geometry while replacing UCB optimism with a DEC-optimized decision rule, achieving best of both worlds. MAMS also optimizes the MAIR/DEC side and further obtains pathwise adversarial robustness, albeit at the cost of losing the GP information-gain geometry. On the other hand, the hub–cloud separation shows that heterogeneous GP–UCB can achieve low hub-complexity regret on structured truths, while full-class DEC and minimax criteria must still pay for high-complexity cloud alternatives. Moreover, existing finite-action optimism-based results leave the optimality debate open. Thus realized algorithmic information, step-uniform minimax coefficients, and class-wide minimax regret capture different aspects of kernel-bandit complexity. Optimality should therefore be assessed by comparing them together with the dynamic trajectory, estimation complexity, and action-set geometry.

Contributions. The paper makes three contributions.

1. We formulate GP–UCB, (GP-)E2D, and MAMS/MEBO in a common MAIR language. The central theorem is stated as an identity: regret equals an estimation-complexity telescope plus a MAIR gradient bracket. This makes rigorous the separation between information gain and regret-to-information coefficient.

2. We show how the same bracket yields three algorithmic readings. Heterogeneous GP–UCB verifies the bracket through self-normalized calibration and optimism; GP–E2D/GP–IDS uses a GP posterior as an estimator and solves a KL or Hellinger DEC decision problem; MAMS/MEBO chooses beliefs and decisions by a robust MAIR saddle point.

3. We compare the three methods along three axes: estimation complexity, coefficient control, and robustness. GP-posterior methods enjoy realized log-determinant estimation complexity, whereas MAMS pays a covering-entropy cost but gains pathwise adversarial robustness. Conversely, GP–E2D and MAMS optimize the coefficient, while GP–UCB only certifies it after the fact.

A hub–cloud example shows that class-wide minimax and DEC certificates can overstate the difficulty faced by a localized algorithmic prior, motivating new criteria for assessing performance.

2 MAIR: one identity and three specializations

2.1 DMSO notation and regret

We use standard DMSO notation (Foster et al., 2021). A model $M \in \mathcal{M}$ specifies, for every decision $\pi \in \Pi$, an expected reward $r_M(\pi)$ and an observation law $P_{M,\pi}$. Let

$$\pi_M \in \operatorname*{argmax}_{\pi \in \Pi} r_M(\pi)$$

be an optimal decision under M. For a randomized decision $p \in \Delta(\Pi)$ define the one-step regret gap

$$\Delta_M(p) := r_M(\pi_M) - \mathbb{E}_{\pi \sim p} r_M(\pi).$$

For an algorithm ALG and a fixed ground truth $M^\star$, the expected cumulative regret is

$$\mathfrak{R}_T(\mathsf{ALG}, M^\star) := \mathbb{E}_{M^\star,\mathsf{ALG}}\left[\sum_{t=1}^{T}\{r_{M^\star}(\pi_{M^\star}) - r_{M^\star}(\pi_t)\}\right]. \tag{1}$$

We also use the Bayesian and minimax risks

$$\mathfrak{R}_T^{\mathrm{Bayes}}(\mathsf{ALG}, \Pi_0) := \mathbb{E}_{M\sim\Pi_0}\mathfrak{R}_T(\mathsf{ALG}, M), \qquad \mathfrak{R}_T^{\mathrm{mm}}(\mathcal{M}) := \inf_{\mathsf{ALG}} \sup_{M\in\mathcal{M}} \mathfrak{R}_T(\mathsf{ALG}, M).$$

The core thesis is that these three risks can have different scales for the same ambient kernel class.

2.2 The exact MAIR regret identity

For a model class $\mathcal{M}$, a reference belief $\rho \in \mathrm{int}(\Delta(\mathcal{M}))$, and an algorithmic belief $\mu \in \Delta(\mathcal{M})$, write

$$\mu_{o|\pi} := \int_{M\in\mathcal{M}} \mu(M) P_{M,\pi}$$

for the predictive observation law. The MAIR objective is

$$\mathsf{MAIR}_{\rho,\eta}(p, \mu) := \mathbb{E}_{M\sim\mu}\Delta_M(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p} I_\mu(M; O \mid \pi) - \frac{1}{\eta}\mathrm{KL}(\mu\|\rho), \tag{2}$$

where

$$I_\mu(M; O \mid \pi) = \mathbb{E}_{M\sim\mu}\,\mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi}).$$

Here, we fix an algorithm ALG and predictive law that sequentially generate

- decision distributions $p_1, \ldots, p_T$, and
- algorithmic beliefs $\mu_1, \ldots, \mu_T$.

The resulting regret bound depends jointly on the decision distributions and the algorithmic belief sequence. We refer to Xu and Zeevi (2025) for the fully measure-theoretic formulation.

Theorem 2.1 (MAIR regret identity; Xu–Zeevi Theorem 7.1)**.** *This is the exact algebraic form underlying Xu and Zeevi (2025, Theorem 7.1). Consider a locally compact model class $\mathcal{M}$ and a fixed ground truth $M^\star \in \mathcal{M}$. Let an arbitrary algorithm produce decision distributions $p_1, \dots, p_T$ and algorithmic beliefs $\mu_1, \dots, \mu_T$. Let $\rho_t \in \operatorname{int}(\Delta(\mathcal{M}))$ be the round-t reference belief, updated by the posterior rule generated from the previous algorithmic belief:*

$$\rho_{t+1}(\cdot) = \mu_t(\cdot \mid \pi_t, O_t).$$

Assume the displayed derivatives and likelihood ratios are well-defined. Then, for every $\eta > 0$,

$$\begin{aligned}\mathfrak{R}_T(\mathsf{ALG}, M^\star) = &\underbrace{\frac{1}{\eta}\mathbb{E}\left[\log \frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)}\right]}_{\text{estimation complexity}} \\ &\underbrace{+ \mathbb{E}\sum_{t=1}^{T}\left\{\mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t) + \langle \nabla_\mu \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t), \delta_{M^\star} - \mu_t\rangle\right\}}_{\text{regret-to-information coefficient / MAIR bracket}}. \qquad (3)\end{aligned}$$

Perspective. Theorem 2.1 is the central identity of the paper. Its significance is that the two terms in (3) are not artifacts of a proof technique: they are the exact posterior telescope and the exact first-order MAIR bracket. The usual regret bound appears only after one upper-bounds the terminal posterior ratio by a prior code length. This is why the paper can compare GP–UCB, GP–E2D, and MAMS by asking separately how each method pays estimation complexity and how each controls the regret-to-information coefficient.

The perspective developed in Xu and Zeevi (2025) is that frequentist sequential learning can be organized around algorithmic beliefs μ, viewed as predictive laws, together with decision rules induced by reference or posterior beliefs ρ, either through posterior sampling or through robust minimax optimization. The exchange-of-decision duality developed in Xu and Zeevi (2025, Section5.2, Lemmas 5.2–5.3), based on exact Nash equilibria and an inexact Danskin theorem, turns the abstract MAIR objective into a variational problem over predictive beliefs and decision distributions. In particular, the gradient-error condition in Xu and Zeevi (2025, Lemma 5.3 and Theorems 3.4 and 7.1) is the key step that connects this variational formulation to regret bounds. Because MAIR works with a fully specified model in the stochastic setting and admits closed-form gradient calculus, its proof is comparatively direct: it mainly consists of a model-posterior telescoping identity together with the MAIR gradient identity. For completeness, we give below a self-contained and simplified version tailored to the present paper.

Proof. We give the self-contained algebraic proof because the identity is the conceptual center of the paper; it follows and simplifies the posterior-ratio calculation in Xu and Zeevi (2025, pp. 57–58). Fix a history H_{t-1}. The algorithm chooses p_t and μ_t, then draws $\pi_t \sim p_t$ and observes $O_t \sim P_{M^\star,\pi_t}$. The reference belief is updated by

$$\rho_{t+1}(M) = \frac{\mu_t(M) dP_{M,\pi_t}(O_t)}{d\mu_{t,o|\pi_t}(O_t)}.$$

Therefore, for the fixed true model,

$$\log \frac{\rho_{t+1}(M^\star)}{\rho_t(M^\star)} = \log \frac{\mu_t(M^\star)}{\rho_t(M^\star)} + \log \frac{dP_{M^\star,\pi_t}}{d\mu_{t,o|\pi_t}}(O_t). \qquad (4)$$

Taking conditional expectation gives

$$\mathbb{E}\left[\log \frac{\rho_{t+1}(M^\star)}{\rho_t(M^\star)} \middle| H_{t-1}\right] = \log \frac{\mu_t(M^\star)}{\rho_t(M^\star)} + \mathbb{E}_{\pi\sim p_t} \mathrm{KL}(P_{M^\star,\pi} \| \mu_{t,o|\pi}). \tag{5}$$

By Lemma 2.3, the MAIR bracket equals

$$\begin{aligned} B_t(M^\star) &:= \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t) + \langle \nabla_\mu \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t), \delta_{M^\star} - \mu_t\rangle \\ &= \Delta_{M^\star}(p_t) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p_t} \mathrm{KL}(P_{M^\star,\pi}\|\mu_{t,o|\pi}) - \frac{1}{\eta}\log\frac{\mu_t(M^\star)}{\rho_t(M^\star)}. \end{aligned}$$

Combining this display with (5) yields the one-step identity

$$\Delta_{M^\star}(p_t) = B_t(M^\star) + \frac{1}{\eta}\mathbb{E}\left[\log \frac{\rho_{t+1}(M^\star)}{\rho_t(M^\star)} \middle| H_{t-1}\right]. \tag{6}$$

Taking expectations and summing over t telescopes the logarithm:

$$\mathfrak{R}_T(\mathsf{ALG}, M^\star) = \mathbb{E}\sum_{t=1}^{T} \Delta_{M^\star}(p_t) = \mathbb{E}\sum_{t=1}^{T} B_t(M^\star) + \frac{1}{\eta}\mathbb{E}\log\frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)}. \tag{7}$$

This is (3). The prior-mass inequality follows from $\rho_{T+1}(M^\star) \le 1$. □

Corollary 2.2 (Deterministic specialization of the MAIR identity: dropping the outer expectation)**.** *Consider the setting of Theorem 2.1. Suppose that, conditional on the realized history H_{t-1}, the algorithm chooses a deterministic action $\pi_t = \pi_t(H_{t-1})$, so that $p_t = \delta_{\pi_t}$. For a fixed environment $M^\star$, pathwise along every realized trajectory,*

$$\mathfrak{R}_T(\mathsf{ALG}, M^\star) = \frac{1}{\eta}\mathbb{E}_{M^\star}\left[\log\frac{\rho_{T+1}(M^\star)}{\rho_1(M^\star)}\right] + \sum_{t=1}^{T}\left\{\mathsf{MAIR}_{\rho_t,\eta}(\delta_{\pi_t},\mu_t) + \langle\nabla_\mu \mathsf{MAIR}_{\rho_t,\eta}(\delta_{\pi_t},\mu_t), \delta_{M^\star} - \mu_t\rangle\right\}. \tag{8}$$

Thus the usual expectation form of Theorem 2.1 takes expectation over the observation process, while the deterministic-action specialization removes the outer expectation over the decision distribution p_t.

Proof. The proof of Theorem 2.1 first establishes the conditional-expectation identity (6). At each round t, both sides of this identity are interpreted after conditioning on the history H_{t-1}. Taking the outer expectation and summing over time then yields the unconditional form (7). For the deterministic-action specialization, it is enough to stop at the conditional step: with $p_t = \delta_{x_t}$, (6) gives the MAIR gradient-bracket identity in (8). □

2.3 The MAIR gradient bracket

Lemma 2.3 (MAIR gradient bracket and canonical specializations)**.** *Assume the observation laws are dominated and the displayed derivatives exist. For fixed p, ρ, μ, up to an additive constant on the probability simplex,*

$$\frac{\partial}{\partial\mu(M)}\mathsf{MAIR}_{\rho,\eta}(p,\mu) = \Delta_M(p) - \frac{1}{\eta}\log\frac{\mu(M)}{\rho(M)} - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi}).$$

Consequently, for every $M^\star$,

$$\begin{aligned}&\mathsf{MAIR}_{\rho,\eta}(p,\mu) + \langle \nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\mu), \delta_{M^\star} - \mu\rangle \\ &\quad = \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\,\mathrm{KL}(P_{M^\star,\pi}\|\mu_{o|\pi}) - \frac{1}{\eta}\log\frac{\mu(M^\star)}{\rho(M^\star)}.\end{aligned} \tag{9}$$

This bracket has three important specializations.

1. *If $\mu = \rho$, the logarithmic correction vanishes and the bracket becomes the KL-DEC offset*

$$\Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\,\mathrm{KL}(P_{M^\star,\pi}\|\mu_{o|\pi}). \tag{10}$$

In Section 4.1 we will show that for GP-UCB,

self-normalized calibration + UCB optimism $\quad\Rightarrow\quad$ MAIR gradient-error condition.

2. *If $\bar\mu \in \operatorname{argmax}_{\mu\in\Delta(\mathcal{M})} \mathsf{MAIR}_{\rho,\eta}(p,\mu)$ is an interior maximizer, then*

$$\mathsf{MAIR}_{\rho,\eta}(p,\bar\mu) + \langle \nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\bar\mu), \delta_{M^\star} - \bar\mu\rangle \le \mathsf{MAIR}_{\rho,\eta}(p,\bar\mu). \tag{11}$$

This is the MAMS/MEBO bracket control.

3. *Let $p_{M,\pi} = dP_{M,\pi}/d\nu_\pi$ and define, for an auxiliary pair $(\bar\pi,\bar o)$,*

$$\frac{d\mu^{1/2}_{\bar\pi,\bar o}}{d\rho}(M) := \frac{\sqrt{p_{M,\bar\pi}(\bar o)}}{\int \sqrt{p_{N,\bar\pi}(\bar o)}\,\rho(dN)}. \tag{12}$$

Let $h^2(P,Q) := 1 - \int\sqrt{dP\,dQ}$. If $\bar\pi\sim p$ and $\bar o \sim P_{M^\star,\bar\pi}$, then

$$\begin{aligned}&\mathbb{E}_{\bar\pi,\bar o}\Big[\mathsf{MAIR}_{\rho,\eta}(p,\mu^{1/2}_{\bar\pi,\bar o}) + \langle \nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\mu^{1/2}_{\bar\pi,\bar o}), \delta_{M^\star} - \mu^{1/2}_{\bar\pi,\bar o}\rangle\Big] \\ &\quad \le \Delta_{M^\star}(p) - \frac{1}{\eta}\mathbb{E}_{\pi\sim p}\mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\pi}, P_{M,\pi}).\end{aligned} \tag{13}$$

Thus the square-root posterior turns the MAIR bracket into a Hellinger E2D/DEC offset.

Remark 2.4. *The notation $\nabla_\mu \mathsf{MAIR}_{\mu,\eta}(p,\mu)$ means the partial gradient in the second argument of $\mathsf{MAIR}_{\rho,\eta}(p,\mu)$, evaluated at $\rho=\mu$. It does not differentiate the reference belief appearing in the first subscript. The notation $\nabla_\mu \mathsf{MAIR}_{\rho,\eta}(p,\hat\mu)$ denotes the partial gradient of $\mathsf{MAIR}_{\rho,\eta}(p,\mu)$ with respect to its second argument, differentiated at $\mu=\hat\mu$. Both notations simplify the full Gateaux-derivative notation used in Theorem 2.1; we use these simpler conventions throughout.*

Proof. The derivative calculation is direct. The derivative of $\mathbb{E}_\mu\Delta_M(p)$ is $\Delta_M(p)$; the derivative of $\mathrm{KL}(\mu\|\rho)$ is $\log(\mu(M)/\rho(M))+1$; and differentiating the predictive mixture in the mutual-information term gives $\mathrm{KL}(P_{M,\pi}\|\mu_{o|\pi})$, up to the same simplex constant. Pairing the derivative with $\delta_{M^\star}-\mu$ cancels the averaged terms in (2) and proves (9).

The case $\mu=\rho$ is immediate from the logarithmic cancellation.

The case $\bar\mu \in \operatorname{argmax}\mathsf{MAIR}$ is from the first-order optimality.

It remains to justify the square-root claim; this is a standard fact for the squared Hellinger distance. For fixed $\bar\pi$, let

$$Z_{\bar\pi}(\bar o) := \int \sqrt{p_{N,\bar\pi}(\bar o)}\,\rho(dN).$$

Then

$$\log \frac{d\mu^{1/2}_{\bar{\pi},\bar{o}}}{d\rho}(M^\star) = \frac{1}{2}\log p_{M^\star,\bar{\pi}}(\bar{o}) - \log Z_{\bar{\pi}}(\bar{o}).$$

Under $\bar{o} \sim P_{M^\star,\bar{\pi}}$, Jensen's inequality gives

$$\begin{aligned}
\mathbb{E}\log \frac{d\mu^{1/2}_{\bar{\pi},\bar{o}}}{d\rho}(M^\star) &= -\mathbb{E}\log \frac{Z_{\bar{\pi}}(\bar{o})}{\sqrt{p_{M^\star,\bar{\pi}}(\bar{o})}} \\
&\geq -\log \int \sqrt{p_{M^\star,\bar{\pi}}(o)}\, Z_{\bar{\pi}}(o) d\nu_{\bar{\pi}}(o) \\
&= -\log\left(1 - \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar{\pi}}, P_{M,\bar{\pi}})\right) \\
&\geq \mathbb{E}_{M\sim\rho} h^2(P_{M^\star,\bar{\pi}}, P_{M,\bar{\pi}}).
\end{aligned}$$

Averaging over $\bar{\pi} \sim p$, substituting into (9), and dropping the nonnegative KL term proves (13). $\square$

A central organizing perspective of this paper is that the GP–UCB proof can be read as a MAIR decomposition. The round-heterogeneous summation of posterior widths, usually controlled by the elliptical-potential or log-determinant argument, plays the role of a telescoping estimation-complexity term: it is the accumulated algorithmic information or fixed-truth estimation loss. The confidence multiplier, by contrast, enters as a per-round upper bound on the MAIR/DEC bracket, i.e., on the coefficient that converts posterior information into regret. This interpretation may be a bit counterintuitive at first sight from the finite-action UCB proof viewpoint: what is often called the "estimation" difficulty in self-normalized martingale concentration becomes, in MAIR language, part of the regret-to-information coefficient; while the elliptical-potential summation becomes the estimation-complexity telescope. This reversal is precisely the model-based view: posterior widths quantify the "estimation complexity" of the *underlying parameter*, as explained excellently in Lattimore (2023). The resulting unification is therefore not merely an algebraic rearrangement of known facts. It reveals a common round-heterogeneous posterior-telescoping structure while separating the distinct estimation and coefficient mechanisms behind GP-UCB, GP-E2D, and MAMS/MEBO.

2.4 IR, DEC, EBO, and MAIR dictionary

Historically, information-ratio methods were algorithmic and decision-space focused, and often appeared in a ratio form: Thompson sampling and IDS are analyzed by ratios of squared one-step regret to posterior information about the optimal action or model (Russo and Van Roy, 2014, 2016). DEC was developed as a model-class offset complexity, comparing a true model to a reference model and optimizing the worst-case regret-estimation tradeoff (Foster et al., 2021, 2023). The boundary is not sharp. Frequentist GP-IDS uses model classes and the true model to define a regret–information ratio, giving an explicit frequentist analogue of the Bayesian information ratio (Kirschner and Krause, 2018). The EBO functional and mirror-descent information-ratio analysis are formulated in an offset form in the frequentist setting (Lattimore and György, 2021); this offset formulation is the source of the MAIR subscript/parenthesis notation and the associated duality analysis in Xu and Zeevi (2025).

We therefore use a functional, rather than doctrinal, naming convention, consistent with Foster et al. (2021, 2022); Xu and Zeevi (2025). When the belief/posterior path and mutual information are central, we use the language of information ratios. When the minimax offset and true-model/reference-model estimation error are central, we use the language of DEC. The MAIR regret identity provides the common structure: it contains a posterior or density-estimation telescope and a bracket that can be read either as an offset model-based information-ratio coefficient or as an extended functional form of the DEC offset.

3 Kernel bandits with heterogeneous algorithmic priors

3.1 Kernel bandits and matrix-valued ridge geometry

Let $\mathcal{X}$ be a locally compact action set, and let $\phi : \mathcal{X} \to \mathbb{R}^d$ be a feature map into a typically *overparameterized* embedding space. A frequentist linear/kernel truth is

$$f^\star(x) = \langle \theta^\star, \phi(x) \rangle,$$

and observations satisfy

$$Y_t = f^\star(x_t) + \varepsilon_t, \tag{14}$$

where x_t is H_{t-1}-measurable and ε_t is conditionally R-sub-Gaussian. Infinite-dimensional RKHS statements follow by the usual limiting argument; the finite-dimensional notation keeps the role of the matrix regularizer transparent. Such an extension is analogous to replacing a bounded RKHS ball with an RKHS ellipsoid through finite-dimensional cutoff and limiting argument.

Fix a self-adjoint positive-semidefinite precision matrix $\Gamma \succeq 0$ and a reference noise level $\lambda > 0$. To avoid domain distractions, the displayed finite-dimensional formulas are written for $\Gamma \succ 0$. The genuinely semidefinite case is obtained either by using Moore–Penrose inverses on range(Γ) or by replacing Γ with $\Gamma + \zeta I$ and sending $\zeta \downarrow 0$ whenever the resulting code length and information gain are finite. The corresponding algorithmic Gaussian reference is

$$\theta \sim N(0, \lambda\Gamma^{-1}), \qquad Y_t \mid \theta, x_t \sim N(\langle \theta, \phi(x_t) \rangle, \lambda). \tag{15}$$

Equivalently, the algorithmic covariance kernel is

$$k_\Gamma(x, x') := \lambda \phi(x)^\top \Gamma^{-1} \phi(x').$$

Only the prior covariance $\lambda\Gamma^{-1}$ and the observation variance λ enter the posterior formulas. If one wants to regard the prior covariance C as fixed independently of the observation noise, set $\Gamma = \lambda \mathsf{C}^{-1}$; then $k_\Gamma(x, x') = \phi(x)^\top \mathsf{C} \phi(x')$. The standard GP–UCB geometry is the scalar special case. A heterogeneous prior corresponds to a non-scalar Γ: large precision means that a direction is algorithmically expensive; small precision means that the direction is favored by the prior. Thus the scalar ridge parameter is not separate from the prior geometry; it is the isotropic instance of the general PSD precision matrix.

Define

$$V_t^\Gamma := \Gamma + \sum_{s=1}^{t} \phi(x_s)\phi(x_s)^\top, \qquad S_t := \sum_{s=1}^{t} \varepsilon_s \phi(x_s).$$

The algorithmic posterior mean and latent variance are

$$\mu_t^\Gamma(x) = \phi(x)^\top (V_t^\Gamma)^{-1} \sum_{s=1}^{t} Y_s \phi(x_s), \qquad \sigma_{t,\Gamma}^2(x) = \lambda \phi(x)^\top (V_t^\Gamma)^{-1} \phi(x). \tag{16}$$

The realized algorithmic information gain is

$$\mathcal{I}_{T,\Gamma} := \frac{1}{2} \log\det \left(I + \Gamma^{-1/2} \Big(\sum_{t=1}^{T} \phi(x_t)\phi(x_t)^\top \Big) \Gamma^{-1/2} \right) = \frac{1}{2} \log\det(I + \lambda^{-1} K_{\Gamma,T}), \tag{17}$$

where $K_{\Gamma,T} = (k_\Gamma(x_i, x_j))_{i,j\le T}$. The one-step increment is

$$g_t^\Gamma := \frac{1}{2}\log\left(1 + \lambda^{-1}\sigma_{t-1,\Gamma}^2(x_t)\right), \qquad \sum_{t=1}^T g_t^\Gamma = \mathcal{I}_{T,\Gamma}.$$

3.2 Posterior-reference information and fixed-truth transfer

Let q_1 be a reference prior over models, q_t the Bayesian posterior after H_{t-1} under a reference observation channel, and

$$Q_t^\pi(\cdot) := \int P_{M,\pi}(\cdot) q_t(dM)$$

the posterior predictive law. The following identity is the bridge from MAIR potentials to GP information gain.

Theorem 3.1 (Posterior-reference density-ratio telescope)**.** *Assume the likelihood ratios below exist. For predictable decisions π_t,*

$$\frac{dq_{T+1}}{dq_1}(M) = \prod_{t=1}^T \frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t). \tag{18}$$

Consequently,

$$\log\frac{dq_{T+1}}{dq_1}(M) = \sum_{t=1}^T \log\frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t).$$

Under the Bayesian joint law $M \sim q_1$ and $O_t \sim P_{M,\pi_t}$,

$$\mathbb{E}\log\frac{dq_{T+1}}{dq_1}(M) = \sum_{t=1}^T \mathbb{E} I_{q_t}(M; O_t \mid H_{t-1}, \pi_t). \tag{19}$$

Proof. Bayes' rule gives

$$\frac{dq_{t+1}}{dq_t}(m) = \frac{dP_{m,\pi_t}}{dQ_t^{\pi_t}}(O_t).$$

Multiplying the Radon–Nikodym derivatives proves (18). Taking logarithms and conditional expectation under $M \sim q_t$ gives

$$\mathbb{E}\left[\log\frac{dP_{M,\pi_t}}{dQ_t^{\pi_t}}(O_t) \mid H_{t-1}\right] = I_{q_t}(M; O_t \mid H_{t-1}, \pi_t),$$

and summing proves (19). ☐

Corollary 3.2 (Gaussian evaluation: log determinant)**.** *For the heterogeneous Gaussian reference* (15)*,*

$$I_{q_t^\Gamma}(\theta; Y_t \mid H_{t-1}, x_t) = \frac{1}{2}\log\left(1 + \lambda^{-1}\sigma_{t-1,\Gamma}^2(x_t)\right),$$

and hence

$$\sum_{t=1}^T I_{q_t^\Gamma}(\theta; Y_t \mid H_{t-1}, x_t) = \mathcal{I}_{T,\Gamma}.$$

Proof. Conditionally on H_{t-1}, the reference posterior gives $f(x_t)$ Gaussian with variance $\sigma_{t-1,\Gamma}^2(x_t)$, while Y_t has predictive variance $\lambda + \sigma_{t-1,\Gamma}^2(x_t)$. Subtracting Gaussian differential entropies gives the one-step formula. The determinant identity follows from the matrix determinant lemma applied sequentially to the posterior variance update. ☐

The fixed-truth frequentist expectation is different. When the posterior ratio is evaluated at a fixed truth, the resulting term is an information density rather than a strict mutual information; nevertheless, it obeys the same telescoping algebra. The next lemma gives the exact transfer bound needed for frequentist GP-UCB.

Lemma 3.3 (Fixed-truth evaluation under a heterogeneous prior)**.** *Let $\theta^\star$ be fixed, let $Y_T = (Y_1, \ldots, Y_T)^\top$, and let Φ_T be the $T \times d$ matrix with rows $\phi(x_t)^\top$. Set $f_T^\star = \Phi_T\theta^\star$ and $K_{\Gamma,T} = \lambda\Phi_T\Gamma^{-1}\Phi_T^\top$. For the Gaussian reference (15), using the canonical likelihood-ratio version of the Radon–Nikodym derivative,*

$$\log \frac{dq_{T+1}^\Gamma}{dq_1^\Gamma}(\theta^\star) = \mathcal{I}_{T,\Gamma} + \frac{1}{2}Y_T^\top(K_{\Gamma,T} + \lambda I_T)^{-1}Y_T - \frac{1}{2\lambda}\|Y_T - f_T^\star\|_2^2. \tag{20}$$

Consequently,

$$\log \frac{dq_{T+1}^\Gamma}{dq_1^\Gamma}(\theta^\star) \leq \mathcal{I}_{T,\Gamma} + \frac{1}{2\lambda}(\theta^\star)^\top\Gamma\theta^\star. \tag{21}$$

The same inequality holds after expectation under any frequentist law for the noise and the algorithmic randomization.

Proof. For the realized design, the reference marginal law of Y_T is $N(0, K_{\Gamma,T} + \lambda I_T)$, whereas the reference likelihood at $\theta^\star$ is $N(f_T^\star, \lambda I_T)$. Taking the log likelihood ratio gives (20). The variational identity

$$Y_T^\top(K_{\Gamma,T} + \lambda I_T)^{-1}Y_T = \inf_{\theta\in\mathbb{R}^d}\left\{\frac{1}{\lambda}\theta^\top\Gamma\theta + \frac{1}{\lambda}\|Y_T - \Phi_T\theta\|_2^2\right\}$$

follows from completing the square in Gaussian ridge regression. Choosing $\theta = \theta^\star$ cancels the noise quadratic in (20) and proves (21). □

Remark 3.4 (Randomized decisions and the telescope)**.** *The fixed-truth telescope in Lemma 3.3 is pathwise conditional on the realized action sequence. If the algorithm is deterministic conditional on the realized history, then the fixed-truth telescope can be applied pathwise to the realized action sequence. In this case, Lemma 3.3 and Corollary 2.2 directly give the deterministic-action MAIR decomposition. If the algorithm uses a randomized decision rule p_t after observing H_{t-1}, the action draw is treated as part of the realized history, and the same posterior-ratio identity holds conditional on that draw. Consequently, in expected-regret bounds the potential term should be averaged over the observation noise and the algorithm's internal randomization. Thus the relevant form is*

$$\mathbb{E}_{f^\star}\left[\log \frac{dq_{T+1}^{\Gamma,\lambda}}{dq_1^\Gamma}(f^\star)\right] \leq \mathbb{E}_{f^\star}[\mathcal{I}_{T,\Gamma}] + \frac{1}{2}\|f^\star\|_\Gamma^2,$$

rather than a deterministic bound on $\mathcal{I}_{T,\Gamma}$ when the action sequence itself is random.

4 One identity, three algorithms

4.1 GP–UCB: posterior estimation plus optimism

GP–UCB is the practical algorithmic-prior method in the triangle. Its predictive belief is the exact GP posterior reference; its decision rule is optimism rather than a DEC saddle point. The key message is therefore not that UCB optimizes the MAIR coefficient, but that self-normalized calibration and optimism verify the MAIR gradient-error condition after the action has been chosen.

For the heterogeneous GP reference, let

$$Q_{t,\Gamma}^{x} := N\big(\mu_{t-1}^{\Gamma}(x), \lambda + \sigma_{t-1,\Gamma}^{2}(x)\big)$$

be the algorithmic noisy predictive law at action x. For a fixed frequentist truth $f^\star$, define the one-step Gaussian surrogate estimation loss

$$\ell_t^{\Gamma}(f^\star) := \mathrm{KL}\Big(N(f^\star(x_t), \lambda) \,\Big\|\, Q_{t,\Gamma}^{x_t}\Big),$$

and the posterior information increment

$$g_t^{\Gamma} := \frac{1}{2}\log\big(1 + \lambda^{-1}\sigma_{t-1,\Gamma}^{2}(x_t)\big). \tag{22}$$

The realized algorithmic information gain is

$$\mathcal{I}_{T,\Gamma} := \sum_{t=1}^{T} g_t^{\Gamma}.$$

Since GP–UCB is deterministic conditional on the realized history, the decision distribution is $p_t = \delta_{x_t}$. We therefore use the the Corollary 2.2, deterministic-action specialization of the MAIR identity and suppress the expectation over decision randomization. For randomized algorithms such as GP–E2D and MAMS below, we use the expectation form of Theorem 2.1, as discussed in Remark 3.4. The deterministic conditional pathwise identity from Corollary 2.2 takes the following form:

$$\sum_{t=1}^{T} r_t = \frac{1}{\eta}\sum_{t=1}^{T} \ell_t^{\Gamma}(f^\star) + \sum_{t=1}^{T}\left\{r_t - \frac{1}{\eta}\ell_t^{\Gamma}(f^\star)\right\}, \tag{23}$$

where the first term is the posterior-reference estimation complexity and the second term is the regret-to-information coefficient, or MAIR bracket. The point of the GP–UCB proof below is that optimism controls the bracket with exactly the correction needed to replace $\sum_t \ell_t^{\Gamma}(f^\star)$ by the posterior information gain $\sum_t g_t^{\Gamma}$.

The heterogeneous-prior GP–UCB rule is

$$x_t \in \operatorname*{argmax}_{x\in\mathcal{X}} \big\{\mu_{t-1}^{\Gamma}(x) + \beta_{t-1}^{\Gamma}\sigma_{t-1,\Gamma}(x)\big\}. \tag{24}$$

Let $x^\star \in \operatorname{argmax}_x f^\star(x)$. The relevant confidence event is

$$\mathcal{E}_T(\delta) := \big\{\forall t \le T, \forall x \in \mathcal{X} : |f^\star(x) - \mu_{t-1}^{\Gamma}(x)| \le \beta_{t-1}^{\Gamma}(\delta)\sigma_{t-1,\Gamma}(x)\big\}. \tag{25}$$

A standard self-normalized argument gives, up to the usual time-uniform logarithmic adjustment,

$$\beta_t^{\Gamma}(\delta) = \frac{\sqrt{(\theta^\star)^\top \Gamma \theta^\star}}{\sqrt{\lambda}} + \frac{R}{\sqrt{\lambda}}\sqrt{2\left(\mathcal{I}_{t,\Gamma} + \log\frac{1}{\delta}\right)}. \tag{26}$$

The first term is the algorithmic code length of the truth under the heterogeneous prior; it is small for hub truths under the hub-favoring prior and large for cloud truths.

Lemma 4.1 (Self-normalized calibration plus UCB optimism). *On the event* (25)*, heterogeneous-prior GP–UCB satisfies*

$$r_t := f^\star(x^\star) - f^\star(x_t) \le 2\beta_{t-1}^{\Gamma}\sigma_{t-1,\Gamma}(x_t).$$

Let $\kappa_\Gamma^2 := \sup_x k_\Gamma(x,x)$ *and assume* $\sigma_{t,\Gamma}^2(x) \le \kappa_\Gamma^2$. *Then*

$$r_t^2 \le 4C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2 g_t^\Gamma, \qquad C_{\lambda,\kappa_\Gamma} := \frac{2\kappa_\Gamma^2}{\log(1+\kappa_\Gamma^2/\lambda)}. \tag{27}$$

Equivalently, for every $\eta > 0$,

$$r_t - \frac{1}{\eta} g_t^\Gamma \le \eta C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2. \tag{28}$$

Consequently, the exact MAIR bracket with fixed-truth estimation loss satisfies

$$r_t - \frac{1}{\eta}\ell_t^\Gamma(f^\star) \le \eta C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2 + \frac{1}{\eta}\left(g_t^\Gamma - \ell_t^\Gamma(f^\star)\right). \tag{29}$$

Proof. On (25),

$$f^\star(x^\star) \le \mu_{t-1}^\Gamma(x^\star) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x^\star) \le \mu_{t-1}^\Gamma(x_t) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x_t),$$

where the second inequality is the UCB choice. Another use of the confidence event gives

$$\mu_{t-1}^\Gamma(x_t) \le f^\star(x_t) + \beta_{t-1}^\Gamma \sigma_{t-1,\Gamma}(x_t).$$

Combining proves the one-step regret bound. For $0 \le u \le \kappa_\Gamma^2$, the function

$$u \mapsto \frac{u}{\frac{1}{2}\log(1+u/\lambda)}$$

is increasing, so

$$u \le C_{\lambda,\kappa_\Gamma} \frac{1}{2}\log(1+u/\lambda).$$

Substitute $u = \sigma_{t-1,\Gamma}^2(x_t)$ and use (22) to obtain (27). The offset form (28) follows from Young's inequality: if $a_t = C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^\Gamma)^2$, then

$$r_t \le 2\sqrt{a_t g_t^\Gamma} \le \eta a_t + \frac{1}{\eta} g_t^\Gamma.$$

Subtracting $\eta^{-1}\ell_t^\Gamma(f^\star)$ from both sides gives (29). □

Theorem 4.2 (Heterogeneous GP–UCB as a MAIR algorithm)**.** *On the confidence event (25), for every* $\eta > 0$,

$$\sum_{t=1}^T r_t \le \frac{\mathcal{I}_{T,\Gamma}}{\eta} + \eta C_{\lambda,\kappa_\Gamma} \sum_{t=1}^T (\beta_{t-1}^\Gamma)^2. \tag{30}$$

Optimizing over η *gives*

$$\sum_{t=1}^T r_t \le 2\sqrt{C_{\lambda,\kappa_\Gamma}\mathcal{I}_{T,\Gamma}\sum_{t=1}^T (\beta_{t-1}^\Gamma)^2} \le 2\beta_T^\Gamma \sqrt{C_{\lambda,\kappa_\Gamma} T \mathcal{I}_{T,\Gamma}}. \tag{31}$$

Thus

self-normalized calibration + *UCB optimism* $\quad\Rightarrow\quad$ *MAIR gradient-error condition.*

Proof. Start from the exact MAIR decomposition (23):

$$\sum_{t=1}^{T} r_t = \frac{1}{\eta}\sum_{t=1}^{T} \ell_t^{\Gamma}(f^\star) + \sum_{t=1}^{T}\left\{ r_t - \frac{1}{\eta}\ell_t^{\Gamma}(f^\star)\right\}.$$

Apply the one-step MAIR bracket bound (29). The fixed-truth estimation losses cancel:

$$\begin{aligned}\sum_{t=1}^{T} r_t &\le \frac{1}{\eta}\sum_{t=1}^{T} \ell_t^{\Gamma}(f^\star) + \sum_{t=1}^{T}\left[\eta C_{\lambda,\kappa_\Gamma}(\beta_{t-1}^{\Gamma})^2 + \frac{1}{\eta}\left(g_t^{\Gamma} - \ell_t^{\Gamma}(f^\star)\right)\right] \\ &= \frac{1}{\eta}\sum_{t=1}^{T} g_t^{\Gamma} + \eta C_{\lambda,\kappa_\Gamma}\sum_{t=1}^{T}(\beta_{t-1}^{\Gamma})^2 \\ &= \frac{\mathcal{I}_{T,\Gamma}}{\eta} + \eta C_{\lambda,\kappa_\Gamma}\sum_{t=1}^{T}(\beta_{t-1}^{\Gamma})^2.\end{aligned}$$

This proves (30). Optimizing the right-hand side over $\eta > 0$ yields the first inequality in (31). The second follows when β_t^{Γ} is nondecreasing. □

This theorem recovers the familiar realized-information proof of GP–UCB, but now through the MAIR identity. The exact identity first separates regret into a posterior-reference estimation term $\sum_t \ell_t^{\Gamma}(f^\star)$ and a MAIR bracket. Self-normalized calibration and UCB optimism then bound the bracket with the correction $g_t^{\Gamma} - \ell_t^{\Gamma}(f^\star)$, causing the fixed-truth estimation loss to cancel and leaving the realized GP information gain $\mathcal{I}_{T,\Gamma}$. Replacing $\mathcal{I}_{T,\Gamma}$ by the maximal information gain of the heterogeneous kernel k_Γ gives the usual class-wide relaxation. Keeping $\mathcal{I}_{T,\Gamma}$ is the algorithmic version. Frequentist GP–IDS (Kirschner and Krause, 2018), by contrast, directly optimizes the regret-to-information ratio term.

4.2 (GP-)E2D: GP estimation plus localized DEC decision-making

E2D and information-directed sampling take the opposite approach to the coefficient side: given a predictive reference, they choose the decision distribution by solving the regret–estimation tradeoff. This subsection states localized variants of the E2D algorithms. We first state the KL version.

Definition 4.3 (KL GP–E2D rule)**.** *At round t, let the GP predictive reference at $x \in \mathcal{X}$ be*

$$Q_t^x = N\left(\mu_{t-1}^{\Gamma}(x), \lambda + (\sigma_{t-1}^{\Gamma}(x))^2\right).$$

For $\gamma > 0$, the KL GP–E2D decision rule chooses

$$p_t \in \operatorname*{argmin}_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{F}_t} \left\{\Delta_f(p) - \gamma\,\mathbb{E}_{x\sim p}\,\mathrm{KL}(N(f(x),\lambda)\,\|\,Q_t^x)\right\}.$$

We denote the value of this saddle problem by

$$\mathrm{dec}_\gamma^{\mathrm{KL}}(\mathcal{F}_t, Q_t) := \inf_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{F}_t} \left\{\Delta_f(p) - \gamma\,\mathbb{E}_{x\sim p}\,\mathrm{KL}(N(f(x),\lambda)\,\|\,Q_t^x)\right\}.$$

This is the GP analogue of the E2D/DEC saddle point and is closely related to frequentist GP–IDS (Kirschner and Krause, 2018).

Theorem 4.4 (KL GP–E2D oracle decomposition)**.** *Assume that $f^\star \in \mathcal{F}_t$ for all t on the calibration event. Let $(p_t)_{t=1}^T$ be chosen by the KL GP–E2D rule in Definition 4.3. Then, on the calibration event,*

$$\Delta_{f^\star}(p_t) \le \mathrm{dec}_\gamma^{\mathrm{KL}}(\mathcal{F}_t, Q_t) + \gamma\, \mathbb{E}_{x\sim p_t} \mathrm{KL}(N(f^\star(x),\lambda)\,\|\,Q_t^x)$$

for every t. Consequently, using Lemma 3.3,

$$\mathfrak{R}_T(\mathsf{KL\ GP-E2D}, M^\star) \le \sum_{t=1}^T \mathbb{E}_{f^\star}\, \mathrm{dec}_\gamma^{\mathrm{KL}}(\mathcal{F}_t, Q_t) + \gamma\left(\mathbb{E}_{f^\star}\mathcal{I}_{T,\Gamma} + \frac{1}{2\lambda}\|\theta^\star\|_\Gamma^2\right).$$

In the infinite-dimensional RKHS formulation, the term $\frac{1}{2\lambda}\|\theta^\star\|_\Gamma^2$ is replaced by the corresponding RKHS code length.

Proof. Since p_t minimizes the KL GP–E2D saddle problem and $f^\star \in \mathcal{F}_t$, the defining inequality gives

$$\Delta_{f^\star}(p_t) - \gamma\mathbb{E}_{x\sim p_t} \mathrm{KL}(N(f^\star(x),\lambda)\,\|\,Q_t^x) \le \mathrm{dec}_\gamma^{\mathrm{KL}}(\mathcal{F}_t, Q_t).$$

Rearranging yields the one-step bound. Summing over t and applying Lemma 3.3 gives the displayed regret bound. □

Thus GP–E2D keeps the same realized log-determinant estimation complexity as GP–UCB, but optimizes a DEC/IDS decision coefficient rather than relying on UCB optimism.

We next state the Hellinger version. It uses the square-root posterior. Since

$$\sqrt{\varphi_\lambda(y - f(x))} \propto \varphi_{2\lambda}(y - f(x))$$

as a function of f, the square-root GP posterior is an ordinary GP posterior with doubled reference noise.

Definition 4.5 (Hellinger GP–E2D rule)**.** *Let $\widetilde{Q}_t$ denote the square-root GP predictive reference, equivalently the GP predictive law obtained with doubled reference noise 2λ. For $\gamma > 0$, define*

$$\mathrm{dec}_\gamma^{\mathsf{H}}(\mathcal{F}_t, \widetilde{Q}_t) := \inf_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{F}_t} \left\{\Delta_f(p) - \gamma\, \mathbb{E}_{x\sim p} h^2\Big(N(f(x),\lambda), \widetilde{Q}_t^x\Big)\right\},$$

where h^2 denotes squared Hellinger distance. A Hellinger GP–E2D rule chooses any minimizer of this display.

Theorem 4.6 (Hellinger GP–E2D oracle decomposition)**.** *Assume that $f^\star \in \mathcal{F}_t$ for all t on the calibration event. Let $(p_t)_{t=1}^T$ be chosen by the Hellinger GP–E2D rule in Definition 4.5. Then*

$$\mathfrak{R}_T(\mathsf{GP-E2D}, M^\star) \le \sum_{t=1}^T \mathbb{E}_{f^\star}\, \mathrm{dec}_\gamma^{\mathsf{H}}(\mathcal{F}_t, \widetilde{Q}_t) + \gamma\left(\mathbb{E}_{f^\star}\mathcal{I}_{T,\Gamma}^{(2\lambda)} + \frac{1}{2\lambda}\|\theta^\star\|_\Gamma^2\right),$$

where

$$\mathcal{I}_{T,\Gamma}^{(2\lambda)} := \frac{1}{2}\log\det\big(I + (2\lambda)^{-1}K_{\Gamma,T}\big).$$

In the infinite-dimensional RKHS formulation, the norm term is again replaced by the corresponding RKHS code length.

Proof. The proof is the same as for Theorem 4.4, with the KL offset replaced by the squared-Hellinger offset and with the square-root posterior bracket (13). The fixed-truth telescope then gives the tempered log-determinant budget $\mathcal{I}_{T,\Gamma}^{(2\lambda)}$. □

The Hellinger version has the same information-plus-DEC architecture as KL GP–E2D, but with a Hellinger decision term and a tempered log-determinant budget. It is not an unconditional improvement: $h^2 \leq \mathrm{KL}$, so the Hellinger offset is technically weaker, although the tempered information budget is smaller.

4.3 MAMS/MEBO: belief optimization and pathwise robustness

MAMS/MEBO chooses beliefs and decisions by optimizing the MAIR bracket itself. Let $\mathcal{M}_\varepsilon$ be a finite discretization of a kernel/RKHS class on a finite action set. For $f \in \mathcal{M}_\varepsilon$ and $x \in \mathcal{X}$, set $r_f(x) = f(x)$ and $P_{f,x} = N(f(x), \lambda)$. Initialize $\rho_1 \in \mathrm{int}(\Delta(\mathcal{M}_\varepsilon))$. At round t, choose a saddle point

$$(p_t, \mu_t) \in \operatorname*{argmin}_{p\in\Delta(\mathcal{X})} \operatorname*{argmax}_{\mu\in\Delta(\mathcal{M}_\varepsilon)} \mathsf{MAIR}_{\rho_t,\eta}(p, \mu), \tag{32}$$

sample $x_t \sim p_t$, observe Y_t, and update

$$\rho_{t+1}(f) = \frac{\mu_t(f)\varphi_\lambda(Y_t - f(x_t))}{\sum_{g\in\mathcal{M}_\varepsilon} \mu_t(g)\varphi_\lambda(Y_t - g(x_t))}. \tag{33}$$

This is the finite-kernel version of the MAMS/MEBO algorithm.

Define the KL-DEC value of the finite class by

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon) := \sup_{\bar{M}\in\mathrm{conv}(\mathcal{M}_\varepsilon)} \inf_{p\in\Delta(\mathcal{X})} \sup_{f\in\mathcal{M}_\varepsilon} \left\{ \Delta_f(p) - \frac{1}{\eta}\mathbb{E}_{x\sim p}\, \mathrm{KL}(P_{f,x} \| P_{\bar{M},x}) \right\}. \tag{34}$$

Theorem 4.7 (Finite and pathwise MAMS/MEBO regret)**.** *If $f^\star \in \mathcal{M}_\varepsilon$, then MAMS satisfies*

$$\mathfrak{R}_T(\mathsf{MAMS}, f^\star) \leq \frac{\log(1/\rho_1(f^\star))}{\eta} + \mathbb{E}\left[\sum_{t=1}^T \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t)\right] \leq \frac{\log(1/\rho_1(f^\star))}{\eta} + T\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon). \tag{35}$$

For a uniform prior, the first term is $\log|\mathcal{M}_\varepsilon|/\eta$.

More generally, let Π be a prior on model paths $\mathbf{M} = (M_1, \ldots, M_T) \in \mathcal{M}^T$, and measure dynamic regret against the per-round optimum of M_t:

$$\mathfrak{R}_T(\mathsf{ALG}, \mathbf{M}) := \mathbb{E}_\mathbf{M} \sum_{t=1}^T \{r^\star_{M_t} - r_{M_t}(A_t)\}.$$

If MAMS is run on the path class using the path posterior and per-round likelihood $P_{M_t,\pi}$, then for every path $\mathbf{M}$,

$$\mathfrak{R}_T(\mathsf{MAMS}, \mathbf{M}) \leq \frac{1}{\eta}\log\frac{1}{\Pi(\mathbf{M})} + \mathbb{E}_\mathbf{M}\sum_{t=1}^T b_t, \tag{36}$$

where b_t is the corresponding MAIR/DEC saddle value. For a standard switching prior and a path with K switches,

$$\log\frac{1}{\Pi(\mathbf{M})} \lesssim (K+1)\log|\mathcal{M}| + K\log T.$$

The adversarial/pathwise part explains what MAMS buys over GP posterior methods: it can handle changing environments by paying path code length. Without a bounded path complexity, dynamic regret against an arbitrary per-round optimum is impossible.

Proof. For the fixed finite-class statement, apply Theorem 2.1. Since μ_t is an interior maximizer of $\mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu)$ over μ, the first-order optimality condition gives

$$\langle \nabla_\mu \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t), \delta_{f^\star} - \mu_t \rangle \le 0.$$

Hence the MAIR bracket is at most $\mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t)$. The identity and the bound $\rho_{T+1}(f^\star) \le 1$ give

$$\mathfrak{R}_T(\mathsf{MAMS}, f^\star) \le \frac{\log(1/\rho_1(f^\star))}{\eta} + \mathbb{E}\sum_{t=1}^{T} \mathsf{MAIR}_{\rho_t,\eta}(p_t,\mu_t).$$

By the minimax choice of p_t and the MAIR-to-DEC comparison in Lemma 2.3, each saddle value is bounded by $\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon)$, giving (35).

For the pathwise/adversarial statement, enlarge the model class to

$$\overline{\mathcal{M}}_T := \mathcal{M}^T.$$

A path $\mathbf{M} = (M_1, \ldots, M_T)$ is treated as a single model whose round-t reward and observation law are those of M_t. The posterior update on paths uses the likelihood P_{M_t,π_t} at round t. Applying Theorem 2.1 to the finite class $\overline{\mathcal{M}}_T$ gives the same identity with $M^\star$ replaced by $\mathbf{M}$. The MAMS saddle point on the path class bounds the bracket by b_t, and the terminal posterior ratio is at most $\log(1/\Pi(\mathbf{M}))$. Therefore

$$\mathfrak{R}_T(\mathsf{MAMS}, \mathbf{M}) \le \frac{1}{\eta}\log\frac{1}{\Pi(\mathbf{M})} + \mathbb{E}_{\mathbf{M}}\sum_{t=1}^{T} b_t.$$

If Π is a standard switching prior, encoding the initial model costs $\log|\mathcal{M}|$, each new segment costs another $\log|\mathcal{M}|$, and the switch times cost at most $K\log T$ up to constants. Hence a path with K switches satisfies

$$\log\frac{1}{\Pi(\mathbf{M})} \lesssim (K+1)\log|\mathcal{M}| + K\log T.$$

□

Corollary 4.8 (Uniform prior and discretization bookkeeping)**.** *Suppose $\mathcal{M}_\varepsilon$ is an ε-net of a larger kernel class in sup norm and every $f^\star$ has a projection $\hat f \in \mathcal{M}_\varepsilon$ with $\|f^\star - \hat f\|_\infty \le \varepsilon$. If, in addition, the observation channels satisfy*

$$\sup_x \mathrm{KL}(P_{f^\star,x} \| P_{\hat f,x}) \le \chi_\varepsilon,$$

then the same proof with a misspecification comparison yields the schematic bound

$$\mathfrak{R}_T(\mathsf{MAMS}, f^\star) \le \frac{\log|\mathcal{M}_\varepsilon|}{\eta} + T\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_\varepsilon) + 2\varepsilon T + \frac{\chi_\varepsilon T}{\eta}. \tag{37}$$

For Gaussian observations with common variance λ, one may take $\chi_\varepsilon = \varepsilon^2/(2\lambda)$.

This is the precise sense in which MAMS/MEBO provides a finite-covering route to kernel bandits. It is robust and DEC-aligned, but the covering term $\log|\mathcal{M}_\varepsilon|$ and the misspecification bookkeeping can be much larger than the log-determinant information accounting used by GP posterior methods.

5 Comparing GP–UCB, GP–E2D, MAMS, and minimax benchmarks

5.1 The triangle: estimation, coefficient, robustness

The clean comparison is three-dimensional. On the estimation side, GP–UCB and GP–E2D use a coherent GP posterior reference and therefore pay realized log-determinant information gain. Finite MAMS/MEBO pays prior mass, covering entropy, or path code length. On the coefficient side, GP–UCB does not solve the MAIR/DEC saddle point; it verifies the coefficient after the fact through calibration and optimism. GP–E2D and MAMS/MEBO optimize the coefficient directly. On the robustness side, MAMS/MEBO can be made pathwise or adversarial by putting a prior on model sequences, while a GP posterior method relies on one coherent algorithmic prior geometry unless it is safeguarded.

	GP–UCB	**GP–E2D / GP–IDS**	**MAMS / MEBO**
Estimation	GP log determinant $\mathcal{I}_{T,\Gamma}$.	GP log determinant $\mathcal{I}_{T,\Gamma}$ or tempered $\mathcal{I}_{T,\Gamma}^{(2\lambda)}$.	Prior mass, covering entropy, or path code length.
Coefficient	Certified post hoc by self-normalized calibration and optimism.	Optimized by KL/Hellinger DEC or IDS.	Optimized by robust MAIR/DEC saddle point.
Robustness	Strong under a good coherent algorithmic prior; weak against misspecified high-complexity directions.	Local/reference dependent; can be safeguarded.	Class-wide and pathwise adversarial robust, but may be pessimistic.

Thus the paper does not claim a generic information-gain advantage of GP–UCB over GP–E2D. Both can use the same posterior-reference telescope. The localized advantage of GP–UCB appears in instances such as the hub–cloud construction, where optimism and a heterogeneous prior avoid spending information on directions that are already algorithmically implausible. The minimax lesson points the other way: GP–E2D and GP–IDS can have better coefficient control than GP–UCB while maintaining the GP information term, suggesting that UCB need not be coefficient-optimal in the most general class-wide sense.

5.2 A safeguarded route: no worse than any, good on structured truths

A single heterogeneous prior cannot have everything for free. If the prior assigns tiny variance to cloud directions, GP–UCB will be excellent on hub truths but may be slow on cloud truths. If the prior assigns large variance to every direction, it is robust but loses the hub-scale advantage. GP–E2D sits between these two extremes: it keeps the GP posterior estimation geometry, as GP–UCB does, but chooses actions through a DEC/IDS decision rule rather than pure optimism. A clean way to express the desired guarantee is therefore to safeguard the practical GP-based algorithms with a robust base algorithm.

Proposition 5.1 (Meta-oracle form of a safeguarded algorithm)**.** *Suppose a bandit-over-bandit master combines three base algorithms: heterogeneous GP–UCB, GP–E2D, and finite-kernel MAMS.*

Assume it guarantees, for every model M, an overhead $\mathfrak{C}_T$ relative to the best base algorithm:

$$\mathfrak{R}_T(\mathsf{Safe}, M) \leq \min\{\mathfrak{R}_T(\mathsf{UCB}_\Gamma, M), \mathfrak{R}_T(\text{GP-E2D}, M), \mathfrak{R}_T(\mathsf{MAMS}, M)\} + \mathfrak{C}_T.$$

Proof. Both claims follow by taking the corresponding term inside the minimum. Standard corralling algorithms provide such regret-to-best-base guarantees for bandit algorithms under the usual feedback assumptions (Agarwal et al., 2017); the present proposition only records the consequence needed for the comparison. □

This proposition gives the right conceptual target. Heterogeneous GP–UCB explains the favorable algorithmic behavior produced by a well-matched prior geometry. GP–E2D keeps the same GP information-gain estimation budget but replaces optimism by a coefficient-aware DEC/IDS decision rule. MAMS supplies the class-wide and adversarial insurance. When the finite MAMS component is tuned to the DEC/minimax rate of the chosen cover, the safeguarded master is not worse than that class-wide benchmark up to the cover error and the master overhead, while still inheriting the best available GP-based bound on structured truths. The main theorems of this paper do not rely on the master; it is included to clarify why the algorithmic-prior view is compatible with, rather than opposed to, minimax robustness.

6 Hub–cloud separation

The hub–cloud perspective builds on the pointwise-complexity separation of Xu (2026). For broader motivation on pointwise and algorithmic complexity, see also Li and Xu (2026); Xu (2026).

Let $\mathcal{X} = H_M \cup C_N$, where $H_M = \{h_0, h_1, \ldots, h_M\}$ is a hub/leaf set and $C_N = \{c_1, \ldots, c_N\}$ is a large cloud. Use independent-arm features, so the ambient kernel is diagonal. Choose a heterogeneous algorithmic covariance

$$k_\Gamma(h,h) = 1 \quad (h \in H_M), \qquad k_\Gamma(c,c) = \varepsilon_C^2 \quad (c \in C_N),$$

with $\varepsilon_C \ll 1$. Equivalently, the prior precision is much larger on cloud coordinates. The GP–UCB acquisition is the ordinary one for this heterogeneous kernel; no additional acquisition penalty is needed.

Consider hub truths $\mathcal{M}_H$ in which one hub arm $h^\star$ has mean $R > 0$, the other hub arms have mean 0, and all cloud arms have mean 0. These truths have small norm in the heterogeneous RKHS because their cloud coordinates vanish. Consider also cloud truths $\mathcal{M}_C$ in which one cloud arm has mean $S > 0$ and all other arms have mean 0. These have norm of order S/ε_C and are high-complexity under the hub-favoring prior.

Theorem 6.1 (Cloud minimax lower bound and DEC certificate)**.** *Let M_j, $j \in [N]$, denote the cloud model in which arm c_j has mean reward $S > 0$ and every other arm has mean reward zero. Pulling arm x under M_j produces an observation distributed as*

$$Y \mid x, M_j \sim N(r_{M_j}(x), \lambda), \qquad \lambda > 0,$$

independently across rounds conditional on the chosen arms. Let $\mathcal{M}_C = \{M_1, \ldots, M_N\}$. Then every adaptive bandit algorithm satisfies, for $T \leq N/4$,

$$\sup_{M \in \mathcal{M}_C} \mathfrak{R}_T(\mathsf{ALG}, M) \geq \frac{3}{4} ST. \tag{38}$$

Consequently, since the display holds for every algorithm,

$$\mathfrak{R}_T^{\mathrm{mm}}(\mathcal{M}_H \cup \mathcal{M}_C) \geq \frac{3}{4} ST.$$

For the DEC side, let M_0 be the null model with zero mean reward and zero observation mean on every arm, and set $\mathcal{M}_{\mathrm{amb}}^0 := \mathcal{M}_H \cup \mathcal{M}_C \cup \{M_0\}$. With the KL-DEC convention in (34),

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_{\mathrm{amb}}^0) \geq S\left(1 - \frac{1}{N}\right) - \frac{S^2}{2\eta\lambda N} \qquad \text{for every } \eta > 0. \tag{39}$$

Equivalently, the reference-specific DEC offset against the null reference M_0 already has the lower bound on the right side of (39). If one insists on the unaugmented class and uses only convex references generated by the cloud models themselves, then the weaker but still cloud-sensitive bound

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_H \cup \mathcal{M}_C) \geq S\left(1 - \frac{1}{N}\right) - \frac{\log N + 2}{\eta N} \tag{40}$$

holds for $N \geq 2$. Thus the class-wide DEC certificate sees the cloud unless the reference or model class is localized away from $\mathcal{M}_C$.

Proof. We first prove the minimax lower bound. Put the uniform prior on the cloud index $J \sim \mathrm{Unif}([N])$ and run the algorithm on the corresponding model M_J. Let

$$\tau := \inf\{t \geq 1 : x_t = c_J\}$$

be the first time the algorithm samples the good cloud arm, with the convention $\tau = \infty$ if this never occurs. Couple this experiment with the null experiment M_0 as follows. Under M_0 all observations are $N(0, \lambda)$. Draw $J \sim \mathrm{Unif}([N])$ independently of the null observations and let $A_T^0 \subseteq [N]$ be the set of cloud indices sampled by the algorithm during the first T rounds of the null experiment. Since $|A_T^0| \leq T$ almost surely,

$$\mathbb{P}_0\{J \in A_T^0\} = \mathbb{E}_0\left[\frac{|A_T^0|}{N}\right] \leq \frac{T}{N}.$$

Before the arm c_J is sampled, the observations under M_J have exactly the same distribution as under the null model along the realized action path. Hence the event that the true experiment hits c_J by time T is contained, under the above coupling, in the event $\{J \in A_T^0\}$, and therefore

$$\mathbb{P}\{\tau \leq T\} \leq \frac{T}{N} \leq \frac{1}{4}.$$

On the event $\{\tau > T\}$ the algorithm never plays the unique optimal arm during the first T rounds, so it suffers regret S in every round. Thus the Bayes regret under the uniform cloud prior is at least

$$\mathbb{E}_J \mathfrak{R}_T(\mathsf{ALG}, M_J) \geq ST\,\mathbb{P}\{\tau > T\} \geq \frac{3}{4} ST.$$

Since a supremum is at least an average, (38) follows.

We next prove the DEC lower bound. Because $M_0 \in \mathcal{M}_{\mathrm{amb}}^0 \subseteq \mathrm{conv}(\mathcal{M}_{\mathrm{amb}}^0)$, the robust KL-DEC is at least the value obtained by fixing the convex reference to be M_0:

$$\mathsf{DEC}_\eta^{\mathrm{KL}}(\mathcal{M}_{\mathrm{amb}}^0) \geq \inf_{p \in \Delta(\mathcal{X})} \sup_{j \in [N]} \left\{ \Delta_{M_j}(p) - \frac{1}{\eta} \mathbb{E}_{x \sim p} \mathrm{KL}(P_{M_j,x} \| P_{M_0,x}) \right\}.$$

Fix any decision distribution $p \in \Delta(\mathcal{X})$ and write $p_j := p(c_j)$. Since $\sum_{j=1}^{N} p_j \leq 1$, there exists $j_p \in [N]$ with $p_{j_p} \leq 1/N$. Under model M_{j_p} the optimal arm is c_{j_p} and has reward S, while every other arm has reward zero. Therefore

$$\Delta_{M_{j_p}}(p) = S(1 - p_{j_p}) \geq S\left(1 - \frac{1}{N}\right).$$

Moreover, the only arm whose observation law differs between M_{j_p} and M_0 is c_{j_p}; for this arm the two laws are $N(S, \lambda)$ and $N(0, \lambda)$. Hence

$$\mathbb{E}_{x\sim p} \operatorname{KL}(P_{M_{j_p},x} \| P_{M_0,x}) = p_{j_p} \frac{S^2}{2\lambda} \leq \frac{S^2}{2\lambda N}.$$

Combining the two displays gives, for this j_p,

$$\Delta_{M_{j_p}}(p) - \frac{1}{\eta} \mathbb{E}_{x\sim p} \operatorname{KL}(P_{M_{j_p},x} \| P_{M_0,x}) \geq S\left(1 - \frac{1}{N}\right) - \frac{S^2}{2\eta\lambda N}.$$

Since the argument holds for every p, taking $\sup_j$ and then $\inf_p$ proves (39).

It remains to prove the unaugmented convex-reference variant. Let $\bar{M}_C$ be the uniform mixture of the cloud models. This is an admissible reference in $\operatorname{conv}(\mathcal{M}_H \cup \mathcal{M}_C)$. Fix again $p \in \Delta(\mathcal{X})$ and choose j_p with $p_{j_p} \leq 1/N$. Under M_{j_p} the regret gap is still at least $S(1 - 1/N)$. We bound the KL term against the mixture reference. At arm c_{j_p}, the mixture observation density contains the true density with weight $1/N$, so

$$\operatorname{KL}(P_{M_{j_p},c_{j_p}} \| P_{\bar{M}_C,c_{j_p}}) \leq \log N.$$

At a different cloud arm c_ℓ, $\ell \neq j_p$, the law under M_{j_p} is the null Gaussian density, while the mixture density contains this null Gaussian component with weight $1 - 1/N$. Hence

$$\operatorname{KL}(P_{M_{j_p},c_\ell} \| P_{\bar{M}_C,c_\ell}) \leq -\log\left(1 - \frac{1}{N}\right) \leq \frac{2}{N},$$

where the last inequality uses $N \geq 2$. Hub arms have zero mean under both M_{j_p} and $\bar{M}_C$, and therefore contribute zero KL. Thus

$$\mathbb{E}_{x\sim p} \operatorname{KL}(P_{M_{j_p},x} \| P_{\bar{M}_C,x}) \leq p_{j_p} \log N + \frac{2}{N} \sum_{\ell \neq j_p} p(c_\ell) \leq \frac{\log N + 2}{N}.$$

Using the reference $\bar{M}_C$ in the outer supremum of the KL-DEC definition and repeating the previous offset argument gives (40). □

The separation is expressed entirely through the GP–UCB prior geometry. The same algorithmic covariance that gives the upper bound suppresses the cloud in the construction. This also reveals the limitation: the algorithm is not designed to be good on cloud truths. Proposition 5.1 describes how to add a MAMS safeguard if one wants both class-wide insurance and hub-scale adaptation.

7 Discussion and conclusion

7.1 DEC, Bellman rank, estimation gap, and hub–cloud separation

This paper uses DMSO in the standard sense: the decision space, model class, observation laws, and rewards are explicitly specified. DEC is powerful in this setting because it is a clean class-wide offset complexity. A phrase such as "bounded Bellman rank" is different. Bellman rank is a representation-relative algebraic certificate depending on value-function classes, roll-in distributions, and Bellman-error witnesses (Jiang et al., 2017). Without fixing the witness, discrepancy, and observable experiment, bounded Bellman rank is not itself a DMSO model class and cannot be assigned an intrinsic DEC. Bellman rank says that there exists a low-dimensional witness structure under which a class is learnable. DEC says that a fixed observable model class has a particular decision-estimation tradeoff. Once a bilinear-class witness is fixed and its estimation complexity is charged, one can ask DEC-style questions (Du et al., 2021). Without that, a rich witness class can make a lower bound non-intrinsic or vacuous.

There are also gaps within clean DMSO classes. Chen et al. (2024) show that DEC-style lower bounds may need estimation-complexity refinements such as fractional covering to recover tight lower bounds in general interactive problems. Work on ridge-bandits and even multi-armed bandits shows another lesson: minimax regret may not be recovered from a scalar one-round coefficient alone; algorithm dynamics and phases matter (Rajaraman et al., 2023; Gu et al., 2025).

The hub–cloud separation here is different. It does not rely on representation ambiguity or on an omitted estimation term; instead, it shows that the separation persists on the DEC side as well. The class is finite, the observations are Gaussian, and the regret loss is ordinary. The full-class minimax/DEC benchmark is large because the class contains a huge cloud of high-complexity alternatives. A heterogeneous-prior GP–UCB rule intentionally refuses to be robust to those alternatives unless evidence accumulates. Its algorithmic MAIR performance on hub truths is therefore much smaller than the full-class certificate.

7.2 What the GP–UCB optimality debate becomes in MAIR language

The classical GP-UCB proof consists of two components. The first is a posterior-potential argument. The second is a self-normalized calibration combined with UCB optimism, which converts posterior uncertainty into regret. In MAIR language, the log determinant is the telescoping estimation term, whereas the confidence multiplier in the self-normalized calibration controls the MAIR bracket.

This separates GP–UCB optimality into three layers. The first is *log-determinant optimality*: whether $\mathcal{I}_{T,\Gamma}$, or its worst-case relaxation γ_T, is the right estimation complexity. The second is *step-uniform coefficient optimality*: whether one can improve the smallest valid adaptive RKHS confidence multiplier, as a *step-uniform* coefficient, to match minimax lower bounds. The third is rigorous *algorithmic trajectory optimality*: whether the actual GP-UCB trajectory pays the right combined cost of estimation complexity and stepwise coefficient accumulation.

The Vakili–Lattimore line is best read as addressing the second layer, with implications for the third. Vakili et al. (2021a) ask for tight online RKHS confidence intervals under adaptive design; in the GP–UCB proof this is exactly the question of how sharp the UCB route to the MAIR coefficient can be. Lattimore (2023) gives lower-bound evidence that fully adaptive confidence certification can carry an intrinsic dimension or information price in nondegenerate settings. This should not be confused with the finite-action regime, where phased or independence-based algorithms can obtain sharper confidence behavior (Auer, 2002; Valko et al., 2013; Li et al., 2024). Building on the mathematical evidence that log-determinant estimation terms can be suboptimal, while often remaining the right realized confidence multiplier, the current paper adds a further insight complementary to the Vakili–Lattimore line on the second layer: since (GP-)E2D is designed precisely to address *step-uniform coefficient optimality*, that criterion need not be the appropriate certificate for GP-UCB's favorable practical performance.

The finite-action regime discussed above, together with the multi-armed case, shows why GP-E2D should not yet be claimed to be minimax-effective without providing new rates or recovering the known ones. It also illustrates how a DEC lower bound can obscure which component of the regret complexity is actually responsible for the difficulty (Chen et al., 2024). In an independent K-armed Gaussian bandit, a model-based UCB decomposition uses the telescope

$$\mathcal{I}_T^{\mathrm{model}} = \frac{1}{2}\sum_{a=1}^{K} \log(1+\lambda^{-1}N_a(T)).$$

Much of the K-dependence is carried by this model-estimation term, while the finite-action confidence multiplier may be only logarithmic in $|\mathcal{X}|$. A DEC/E2D decomposition can place the same difficulty differently: the DEC offset contains a K-type regret-to-estimation obstruction (Foster et al., 2021). If this coefficient is combined naively with a full model-based information telescope that already pays for all K arms, the analysis can double count the finite-action difficulty. Thus GP–E2D optimizes the coefficient for a chosen offset, but optimality requires compatibility between the estimation telescope and the coefficient benchmark.

7.3 Conclusion

The paper's view is deliberately pluralistic. GP–UCB is a central practical kernel-bandit algorithm because it uses a tractable posterior geometry and often follows trajectories with favorable realized information. E2D and MAMS/MEBO explain what GP–UCB does not explicitly optimize: the regret-to-information coefficient. The MAIR identity puts these perspectives in one equation. It separates the posterior telescope, or estimation complexity, from the MAIR bracket, or regret-to-information coefficient, and thereby clarifies why different algorithms can be optimal for different parts of the same problem.

This separation also informs how one should read the kernel-bandits optimality debate. A bound based only on maximal information gain hides the realized trajectory. A DEC lower bound captures robust class-wide decision difficulty, but it may be driven by parts of the class that a biased algorithmic prior or a finite-action geometry intentionally suppresses. GP–UCB is strong when the posterior geometry is well matched to the truth; GP–E2D keeps this GP estimation oracle while optimizing a local DEC/IDS coefficient; and MAMS/MEBO supplies the robust class-wide and pathwise benchmark. The right question is not simply which method dominates, but where the complexity is paid: in the telescope, in the coefficient, or jointly along the realized algorithmic trajectory. Kernel and GP bandit theory provide a clean setting to make these distinctions mathematically visible.

A Outlook from kernel bandits

The discussion above suggests several questions.

First, there is the finite-action regime in kernel bandits. A promising route is to combine an AMS/EBO-style decision rule with a Gaussian-process posterior reference, so that the estimation side retains the marginal GP log-determinant geometry while the decision side optimizes an AIR/DEC bracket. One possible technical tool is a Gaussian renormalization/Schur-complement argument, in the spirit of Xu (2026, Section A.2), which would replace the full infinite-dimensional posterior by the finite marginal posterior on the queried action set. If such a reduction can be made sharp, an RG-UCB, RG-AMS, or RG-E2D algorithm could apply to the finite-action regime without paying a full RKHS covering-entropy cost. Since Vakili et al. (2021a) emphasizes that existing finite-action kernel-bandit algorithms rely on sophisticated techniques, such an RG-guided algorithm would provide a conceptually simple and potentially useful new approach.

Second, DEC and E2D for an RKHS ball should be interpreted with some caution. For the full RKHS ball, even after cutoff and localization, the DEC calculation does not appear to improve the maximal-information-gain scale in general. Thus a naive GP-E2D implementation may lose the realized log-determinant geometry that makes GP-UCB effective. Kernel bandits therefore provide a clean setting in which DEC optimization can worsen the classical UCB analysis/performance for the RKHS ball, even though DEC remains closely connected to minimax lower-bound methods. This should not be read as saying that DEC is ineffective. Rather, DEC is better viewed as a powerful interactive lower-bound method, closer in spirit to Le Cam-type lower-bound constructions in statistical learning, whose sharpness depends on problem-specific localization and calculation. At present, its most promising role in kernel bandits may be to identify new minimax structures in geometries more irregular than a standard RKHS ball. In this sense, like other lower-bound methods, DEC is not automatically a tight complexity measure by itself; it is a forward-looking framework for simplifying and organizing interactive lower-bound and algorithm-design arguments. Whether it can practically simplify the discovery of previously unknown minimax rates remains to be tested instance by instance.

Third, the minimax viewpoint in kernel regimes (and perhaps any overparameterized regimes) resembles the Bayesian viewpoint more than it may first appear. An RKHS ball plays a role analogous to a Gaussian prior, while an RKHS ellipsoid corresponds to a heterogeneous Gaussian prior. The difference is partly one of geometry: whether one regards the ball constraint or the Gaussian covariance structure as the more natural description of prior information. In this regime, maximal information gain behaves like an effective dimension with a cutoff, and the RKHS norm specifies where the cutoff occurs. Thus kernel bandits resemble the hub–cloud and Bayesian overparameterized settings: the ambient class may be infinite-dimensional, but the relevant complexity is selected by a localized geometry. In finite-dimensional linear models this distinction is less visible; in overparameterized kernel classes it becomes central. As emphasized in Li and Xu (2026), the RKHS norm can be viewed as a cutoff that prevents the dimension from becoming infinite, while pushing the remaining directions into approximation error. This raises a basic question: if DEC/E2D worsens the UCB rate for the standard RKHS ball and kernel classes, where can it offer genuine improvement? A natural answer is that improvement may come from more heterogeneous geometry and kernels, but the choice of such geometry is itself part of the modeling and localization problem.

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